Assignment 1 IPv4 Addressing Foundations

Networking Fundamentals

Assessment	Individual assignment	**Total**	50 marks
Suggested time	60 minutes	**Due date**	As posted in Moodle
Submission	One PDF or DOCX	**Topic**	Classful IPv4 addressing
Filename	Surname_Firstname_A1	**Attempts**	Follow Moodle settings

Purpose

Complete this assignment to show that you can recognize IPv4 address classes, apply default subnet masks, identify network and broadcast addresses, and check whether a default gateway belongs to the correct network.

Instructions

- Answer every question and show your working where a calculation is required.
- Use the classful A, B, and C rules taught in the lesson. Do not apply custom CIDR or VLSM subnetting in this assignment.
- Network and broadcast addresses cannot be assigned to devices.
- You may use a calculator, but your submitted answers must be your own work.
- Type your answers in this document or write clearly on a printed copy.
- Submit one PDF or DOCX file through the Moodle assignment link by the posted deadline.

Student Details

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Class group	TEAM-05	**Submission date**	28/09/2026

Mark Allocation

Section	What is assessed	Marks
Part A	Core concepts	10
Part B	Classful address identification	15
Part C	Network and broadcast analysis	15
Part D	Applied troubleshooting	10
	Total	**50**

Part A Core Concepts

Answer each question in one to four sentences. Each question is worth 2 marks.

1 Define an IPv4 address and explain why the address must be unique on the same network. **[2 marks]**

An IPv4 address is a unique 32-bit numerical label assigned to any device connected to a network that uses the Internet Protocol for communication.

Routers and switches use destination IP addresses to forward data packets to the correct recipient. If two devices share the same address, the network cannot determine which device should receive the data

2 State one difference between flat and hierarchical addressing. Give one example of each type. **[2 marks]**

Flat addressing is used in intranetwork communication which is assigned statically. While hierarchical addressing is used in internetwork communication which is assigned dynamically.

Flat addressing example is personal name or MAC address. Hierarchical addressing example is zip code or IP address

3 Compare static and dynamic IP addressing. Name the protocol normally used to assign an address dynamically. **[2 marks]**

Static IP address is a manually assigned IP address to a device by an administrator and this IP address is constant and does not change.

Dynamic IP address is assigned to device each time it starts. It Requires less human intervention

Protocol used is Dynamic Host Configuration Protocol (DHCP)

4 State one difference between classful and classless addressing. What does CIDR stand for? **[2 marks]**

Classful addressing uses fixed, predefined classes (A, B, C, D, and E) with implicit subnet masks, whereas classless addressing uses Classless Inter-Domain Routing (CIDR) with flexible, explicit variable-length subnet masks (/24)

CIDR - Classless Inter-Domain Routing

5 List the four IPv4 configuration items presented in the lesson for normal name-based Internet access. **[2 marks]**

1) IP Address
2) Broadcast Address
3) Subnet Mask
4) Default Gateway

Part B Classful Address Identification

For each address, identify the class, write the network and host pattern, and give the default subnet mask. Each row is worth 3 marks.

IPv4 address	Class	Network and host pattern	Default subnet mask
23.14.6.9	A	N.H.H.H	255.0.0.0
145.80.12.200	B	N.N.H.H	255.255.0.0
198.51.100.27	C	N.N.N.H	255.255.255.0
9.200.10.1	A	N.H.H.H	255.0.0.0
188.44.9.72	B	N.N.H.H	255.255.0.0

1. 23.14.6.9 The first octet is **23**. Class A range is **1 to 126**. Therefore: Class = **A** Pattern = **N.H.H.H** Default mask = **255.0.0.0**	**2. 145.80.12.200** The first octet is **145**. Class B range is **128 to 191**. Therefore: Class = **B** Pattern = **N.N.H.H** Default mask = **255.255.0.0**	**3. 198.51.100.27** The first octet is **198**. Class C range is **192 to 223**. Therefore: Class = **C** Pattern = **N.N.N.H** Default mask = **255.255.255.0**
4. 9.200.10.1 The first octet is **9**. Class A range is **1 to 126**. Therefore: Class = **A** Pattern = **N.H.H.H** Default mask = **255.0.0.0**	**5. 188.44.9.72** The first octet is **188**. Class B range is **128 to 191**. Therefore: Class = **B** Pattern = **N.N.H.H** Default mask = **255.255.0.0**	

Part C Network and Broadcast Analysis

Use the default classful mask for each address. Complete all five requested values. Each address is worth 5 marks.

1 34.72.6.19

Network address	34.0.0.0	**Broadcast address**	34.255.255.255
First usable host	34.0.0.1	**Last usable host**	34.255.255.254

Maximum usable hosts	**Calculation:** Host bits = 24 Maximum hosts = $2^{24} - 2$ = 16,777,216 − 2 = **16,777,214**

2 150.20.7.200

Network address	150.20.0.0	**Broadcast address**	150.20.255.255
First usable host	150.20.0.1	**Last usable host**	150.20.255.254
Maximum usable hosts	**Calculation:** Host bits = 16 Maximum hosts = $2^{16} - 2$ = 65,536 − 2 = **65,534**		

3 203.0.113.44

Network address	203.0.113.0	**Broadcast address**	203.0.113.255
First usable host	203.0.113.1	**Last usable host**	203.0.113.254
Maximum usable hosts	**Calculation:** Host bits = 8 Maximum hosts = $2^{8} - 2$ = 256 − 2 = **254**		

Part D Applied Troubleshooting

Read the scenario, then answer all five questions.

A workstation in a training room has the following IPv4 settings. The workstation can communicate with some local devices but cannot reach resources on another network.

Setting	Configured value
IPv4 address	201.110.213.28
Subnet mask	255.255.255.0
Default gateway	201.110.214.1
DNS server	8.8.8.8

1 Give the network address and broadcast address for the workstation. **[2 marks]**

Network address: **201.110.213.0**
Broadcast address: **201.110.213.255**

2 Is the configured default gateway valid for this workstation? Explain your answer. **[2 marks]**

No, the configured default gateway is not valid.

The workstation belongs to the network: **201.110.213.0/24.** A valid gateway must be in the same network as the workstation. The gateway **201.110.214.1** belongs to a different network: **201.110.214.0/24**

Therefore, it cannot serve as the workstation's default gateway.

3 Give one valid default gateway address. Explain why the network and broadcast addresses cannot be used as the gateway. **[2 marks]**

A valid gateway could be: **201.110.213.1**

The network is **201.110.213.0** and the broadcast address is **201.110.213.255**.

Therefore, usable host addresses range from: **201.110.213.1 to 201.110.213.254**

The network address **201.110.213.0** cannot be assigned to a device, and the broadcast address **201.110.213.255** cannot be assigned either.

So **201.110.213.1** is a valid default gateway.

4 With the incorrect gateway, what communication may still work and what communication is likely to fail? **[2 marks]**

Communication with devices **on the same local network** may still work because they are directly reachable within the **201.110.213.0/24** network.

However, communication with **devices on other networks**, including the Internet, is likely to fail because the workstation has an invalid default gateway.
Local network communication may work, but communication with remote networks and the Internet will likely fail.

5 A lab contains 25 student computers. Recommend static or dynamic addressing for the computers and name the service that would provide the addresses. **[2 marks]**

I would recommend **dynamic addressing using DHCP**. The service required is a **DHCP server**.
Use **dynamic IP addressing** provided by a **DHCP server**.

DHCP automatically assigns IP addresses, subnet masks, default gateways and other network settings to the 25 computers, reducing manual configuration errors.